

Catalyst Reaction Screener for TradingView

A practical guide to what the script does, what inputs it takes, why it exists, when to run it, and how to interpret the output in Pine Screener.

Bottom line

This indicator is a catalyst-reaction screener, not a headline scraper. It is designed to surface stocks that are acting like they have a meaningful catalyst right now by ranking gap size, premarket volume, premarket range, range position, and—when available—fresh earnings plus EPS surprise.

At a glance		Why it exists	Main output
Watchlist scan		Pine Screener cannot directly read arbitrary news headlines inside the script, so the indicator measures the market footprint of a catalyst instead.	Sortable columns: SetupScore, Direction, GapPct, PMVolPctADV, PMRangePctATR, PosInPmRange, FreshEarnings, EPSSurprisePct, DollarVol20M.

Script covered: Catalyst Reaction Screener [Pine Screener]

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1. Executive Summary

This script is a TradingView-native indicator built for Pine Screener. Pine Screener scans a watchlist with one selected indicator, then exposes that indicator's plots and alert conditions as sortable and filterable outputs.[1][2] The script was written specifically to fit inside that model.

Its purpose is to answer a practical trading question: which stocks are reacting strongly enough right now that they deserve immediate chart attention? Instead of trying to read the news text itself, the script looks for the footprint that an important catalyst usually leaves behind—an outsized move away from the prior close, unusually strong premarket participation, a meaningful premarket range, and price holding near the strong side of that range.

Earnings are the only explicit catalyst family this script can quantify directly inside Pine because TradingView provides structured earnings data through `request.earnings()`. Everything else—FDA decisions, SEC filings, contract wins, analyst notes, macro shocks—must still be confirmed manually after the screener identifies the reacting name.[3]

What the script is

- A premarket and early-session ranking tool for liquid U.S. stocks.
- A fast way to sort a watchlist by reaction quality instead of manually opening dozens of charts.
- A workflow bridge between News/Filings and actual trade selection.

What the script is not

- It is not a general web scraper or a live Reuters/FDA/SEC parser.
- It is not a strategy that sends or simulates orders.
- It is not a replacement for validating the actual cause of the move.
- It is not a market-wide universe scanner; it only evaluates the watchlist you select in Pine Screener.[1]

Design constraint that matters

TradingView currently limits Pine Screener to one indicator per screen, only specific timeframes, no more than five separate `request.*()` calls in the scanned script, and calculations on only the last 500 bars. This indicator intentionally stays inside those boundaries.[1]

2. How the Script Works

The script follows a six-step pipeline. Each symbol in the screener is processed independently. The goal is not to predict catalysts; it is to measure whether the current tape behaves like a real catalyst is in play.

Step	Stage	What happens
1	Build an extended-hours context	Creates an extended-session ticker with <code>ticker.modify(syminfo.tickerid, session.extended)</code> so the script can read premarket bars when the feed supports extended data.[4][5]
2	Track premarket state	Inside that extended context, the function <code>f_pm_state()</code> accumulates today's premarket high, premarket low, and premarket volume during the session window defined by <code>pmSession</code> and <code>tz</code> .
3	Pull daily reference values	Requests the previous daily close, the 20-day average daily volume, and the 14-day daily ATR from the regular ticker. These numbers become the baseline for gap size, normalized participation, and volatility context.
4	Pull earnings data	Uses <code>request.earnings()</code> to retrieve actual EPS and estimated EPS. TradingView documents that <code>request.earnings()</code> can return actual and estimate fields and that the values reflect the data as it comes in.[3]
5	Apply candidate filters	The script determines whether a symbol is liquid enough, moving enough, and participating enough to count as a serious reaction candidate.

Step	Stage	What happens
6	Score and classify	If the symbol passes the filters, the script assigns a long score or short score, sets Direction to 1, -1, or 0, and exposes all key metrics as screener columns and alert conditions.

Core calculations in plain English

Metric	Formula	Why it matters
gapPct	$(\text{lastPrice} / \text{prevClose} - 1) * 100$	How far price is from yesterday's official close.
pmVolPctAdv	$(\text{pmVol} / \text{adv20}) * 100$	How much of a normal full day's volume has already traded in premarket.
pmRangePctAtr	$((\text{pmHigh} - \text{pmLow}) / \text{atr14d}) * 100$	How large the premarket range is relative to the stock's usual daily volatility.
posInPmRange	$100 * (\text{lastPrice} - \text{pmLow}) / (\text{pmHigh} - \text{pmLow})$	Where price is sitting inside the premarket range: near 100 = holding high; near 0 = holding low.
epsSurpriseRaw	$((\text{epsActual} - \text{epsEstimate}) / \text{abs}(\text{epsEstimate})) * 100$	Percent EPS surprise on the earnings event bar when estimate data exists.

This is what makes the indicator useful: gapPct tells you magnitude, pmVolPctAdv tells you participation, pmRangePctAtr tells you whether the move is large relative to the stock's own history, and posInPmRange tells you whether the move is being accepted or faded.

Candidate logic

Long candidate

A symbol becomes a long candidate only if it is a stock, passes the liquidity filters, passes the reaction filters, passes the catalyst gate, has a positive gap, and is trading in the upper portion of its premarket range.

Short candidate

A symbol becomes a short candidate only if it is a stock, passes the liquidity filters, passes the reaction filters, passes the catalyst gate, has a negative gap, and is trading in the lower portion of its premarket range.

The score is heuristic, not magical. The script weights larger gaps more heavily, then adds normalized premarket volume, adds range relative to ATR, adds a bonus for fresh earnings, adds a capped EPS surprise contribution when available, and finally adds a bonus for holding the strong side of the premarket range. The result is a ranking number called SetupScore.

3. Inputs: What the Script Takes and How They Change Outcomes

There are no mandatory external data inputs beyond selecting a watchlist and a supported screener timeframe. Every script input below has a default value. In practice, this means you can paste the code, save it, open Pine Screener, and run it immediately.

Where people most often change the behavior is not in the formulas—it is in the thresholds. Higher thresholds produce fewer, cleaner names; lower thresholds produce more names but with more noise.

Input	Default	What it controls	How changing it affects results
tz	"America/New_York"	Reference timezone for the session window.	If changed incorrectly, the premarket window can shift and distort all PM metrics.
pmSession	"0400-0930"	The time window counted as premarket.	Widening it includes earlier prints and more volume; narrowing it creates a stricter read.
freshWindowBars	156	How long an earnings event remains "fresh."	Higher values make more earnings names count as fresh; lower values make the earnings filter stricter. This input is timeframe-sensitive.
minPrice	5.0	Minimum share price allowed.	Raise it to avoid low-priced names; lower it to admit smaller or more speculative stocks.
minDollarVol20M	20.0	Minimum 20-day average dollar volume, in millions.	Raise it for better liquidity and cleaner execution; lower it for more mid/small-cap exposure.
minAbsGapPct	2.0	Minimum absolute gap from the prior close.	Raise it to focus on larger catalyst reactions; lower it to catch more subtle moves.
minPmVolPctAdv	5.0	Minimum premarket volume as a percent of 20-day ADV.	Raise it to require stronger participation; lower it to permit thinner movers.
minLongRangePos	60.0	Minimum position inside PM range for long setups.	Raise it to require stronger hold near the PM high; lower it to catch earlier bases.
maxShortRangePos	40.0	Maximum position inside PM range for short setups.	Lower it to require stronger hold near the PM low; raise it to allow more rebound risk.
requireFreshEarnings	false	When true, only fresh earnings names can pass the catalyst gate.	Turn it on for earnings-only scans. Leave it off for broad catalyst-reaction scans.

Timeframe calibration matters

The biggest hidden detail is freshWindowBars. The default of 156 bars makes sense on 5-minute scans, where it represents roughly one full U.S. premarket-plus-cash-session span. If you move to 15-minute scans, 156 bars stretches much farther in clock time. That can make "fresh earnings" less fresh than you intended.

If you scan on	Use freshWindowBars around	Why
1-minute	780	About 13 hours
5-minute	156	About 13 hours
15-minute	52	About 13 hours
30-minute	26	About 13 hours
60-minute	13	About 13 hours

Practical recommendation

Use 5-minute scans as the default configuration. The script's comments and thresholds were written with that workflow in mind.

4. Internal Variables and Output Columns

You asked for variables and outputs explicitly, so this section separates internal working values from what Pine Screener actually displays.

Internal variable	Meaning	Why it matters
extTicker	Extended-session version of the current symbol.	Makes premarket measurements possible when extended data is available.[4][5]
extClose	Latest close from the extended context.	Acts as the current price for gap calculations.
pmHigh / pmLow / pmVol	Today's running premarket high, low, and accumulated volume.	They reset on a new day.
prevClose	Yesterday's daily close.	The anchor for gapPct and dollar-volume normalization.
adv20	20-day average daily volume.	Used to express premarket participation in normalized terms.
atr14d	14-day daily ATR.	Used to measure whether the premarket range is large relative to the stock's normal behavior.
epsActual / epsEstimate	Most recent earnings actual and estimate when available.	Used to mark earnings events and compute EPS surprise.[3]
latestEpsSurprise	The most recent non-na earnings surprise value.	Persists the last known surprise for screener use after the event bar.

Column	Meaning	How to use it
SetupScore	Primary ranking value.	Sort descending to bring the strongest reaction candidates to the top.
Direction	1 = long candidate, -1 = short candidate, 0 = ignore.	Use it as the first coarse filter before reading the rest of the table.
GapPct	Percent move from prior close.	Shows direction and size of the reaction.
AbsGapPct	Absolute value of GapPct.	Useful when you only care that the move is big, not the sign.
PMVolPctADV	Premarket volume as a percent of 20-day ADV.	One of the best quick tests for whether the move is attracting real participation.
PMRangePctATR	Premarket range as a percent of daily ATR.	Helps distinguish a real expansion day from a small wobble.
PosInPmRange	Current position within the premarket range, 0 to 100.	Higher is stronger for longs; lower is stronger for shorts.
FreshEarnings	1 if the latest earnings event is inside the chosen freshness window, else 0.	Useful as a filter during earnings season.
EPSSurprisePct	Latest computed EPS surprise percentage, if available.	Adds context to earnings-driven names; can be blank for non-earnings names.
DollarVol20M	20-day average dollar volume in millions.	A fast liquidity proxy for avoiding thin names.

Alerts

The script defines two `alertcondition()` triggers: Long Candidate and Short Candidate. TradingView's documentation is explicit that `alertcondition()` makes triggers available, but users still need to create the actual alert from the Create Alert dialog in the platform.[6]

That means the script can tell TradingView when a condition is true, but it does not automatically create or arm alerts by itself.

5. When to Run the Script and What to Expect

Best workflow: premarket scan

Run the screener on a 5-minute timeframe from roughly 8:00 a.m. ET until a few minutes before the open. By that point, enough premarket volume has typically accumulated to make PMVolPctADV meaningful, and enough price discovery has occurred to make pmHigh, pmLow, and PosInPmRange useful.

Secondary workflow: after-open continuation scan

Run the screener on 15-minute bars after 9:35 a.m. ET if your goal is to find names that are holding or failing their premarket structure after the bell. When you do this, adjust freshWindowBars downward so that FreshEarnings continues to mean what you think it means.

Earnings-only mode

Turn requireFreshEarnings on when you want the screener to behave like an earnings reaction filter. This will usually reduce the result count but improve the odds that the strongest names are reacting to a specific earnings event rather than unrelated news.

Broad catalyst mode

Leave requireFreshEarnings off when you want a broader catalyst-reaction scanner. In that mode the script will surface strong movers regardless of whether Pine can tag a fresh earnings event. This is the right setting for macro shocks, FDA headlines, SEC filings, contract wins, analyst notes, or sector-wide sympathy moves—but the user still needs to verify the cause manually.

What to expect	Typical behavior
If the day is quiet	Most symbols will have Direction = 0, low SetupScore, or weak premarket participation.
If it is an active earnings day	A few names should cluster at the top with high SetupScore, high PMVolPctADV, and either positive or negative EPS surprise.
If it is a macro or sector shock day	Multiple related names may appear even without FreshEarnings = 1.
If extended-hours data is unavailable for a symbol	Premarket metrics may be blank or too weak, and the symbol may fail the reaction gate.

How to launch it in TradingView

- Paste the script into Pine Editor, save it, and add it to Favorites.
- Open Products → Screeners → Pine.
- Choose the watchlist you want to scan. Pine Screener works from watchlists, not from the entire market universe.[1]
- Select this indicator as the screener study.
- Use a supported screener timeframe such as 5m or 15m.[1]
- Sort by SetupScore and filter by Direction, GapPct, PMVolPctADV, or FreshEarnings as needed.

6. How to Read the Output Like a Trader

The most important thing to understand is that SetupScore is a triage number, not a verdict. It answers: which names deserve immediate attention first? It does not answer: which name is guaranteed to continue?

Pattern in the table	Interpretation
High SetupScore + Direction 1 + PosInPmRange > 80	Strong long-bias candidate. The move is not only large; it is holding near the strong side of the premarket range.
High SetupScore + Direction -1 + PosInPmRange	Strong short-bias candidate. The move is holding near the weak side of the range.

Pattern in the table	Interpretation
< 20	
High gap but low PMVolPctADV	Be careful. The move may be real, but participation has not yet confirmed it.
High PMVolPctADV but small gap	There is attention, but not yet enough price displacement to stand out.
FreshEarnings = 1 and large EPSSurprisePct	Useful confirmation that an earnings event is likely a meaningful driver rather than noise.
FreshEarnings = 0 with strong score	Still tradable, but the catalyst is probably non-earnings and needs manual confirmation.

Fast reading rule

Sort by SetupScore, ignore Direction = 0, then read GapPct, PMVolPctADV, and PosInPmRange together. If all three agree, the symbol is worth opening on a chart immediately.

7. Limitations, Edge Cases, and Caveats

- The script cannot directly read arbitrary headlines or filings from the open web. That is an inference from TradingView's documented request families: the platform exposes structured request.*() datasets such as security, earnings, financial, economic, and seed data—not a generic web request for news text.[3]
- The script explicitly filters to syminfo.type == "stock", so ETFs, many depositary receipts, and non-stock instruments will not qualify even if they are in the watchlist.
- Extended-hours behavior depends on symbol and data feed support. TradingView notes that extended sessions only apply when the symbol and the user's feed allow for them.[5]
- SetupScore is a handcrafted ranking heuristic. It is excellent for prioritization, but it is not a statistically validated expectancy model by itself.
- FreshEarnings only means recent according to your bar count and timeframe. If you change timeframes and do not recalibrate freshWindowBars, your results can drift.
- Pine Screener only calculates on the last 500 bars and supports only a limited set of timeframes.[1] That is usually fine for this script, but it still matters when you try to stretch settings too far.
- General Pine scripts are also subject to broader request.*() limits—40 unique calls, or 64 on Ultimate plans—but this script is well below that ceiling and uses only four separate request families.[7]

8. Suggested Presets

Preset	Suggested settings	Trade-off
Conservative large-cap scan	minPrice = 10, minDollarVol20M = 100, minAbsGapPct = 2.5, minPmVolPctAdv = 8, minLongRangePos = 70, maxShortRangePos = 30	Fewer names, better liquidity, stronger confirmation.
Balanced default scan	Keep defaults	Good general-purpose setting for liquid U.S. stock watchlists.
Aggressive mid-cap scan	minPrice = 3, minDollarVol20M = 10, minAbsGapPct = 1.5, minPmVolPctAdv = 3, minLongRangePos = 55, maxShortRangePos = 45	More names and earlier signals, but noisier and harder to execute cleanly.
Earnings-only scan	requireFreshEarnings = true; keep 5m timeframe; tune freshWindowBars to timeframe	Best around earnings season when you want explicit event confirmation.

As a rule: raise liquidity and participation thresholds when you want cleaner names; lower them only when you deliberately want more speculative exposure.

9. Final Takeaway

The script's job is not to tell you the story. Its job is to tell you which symbols are acting like the story matters. In practice, that is exactly what a good premarket screener should do inside TradingView: narrow the field fast, rank the best candidates, and hand you a short list worth validating and trading.

Use it as a first-pass reaction engine. Then open the chart, confirm the headline, define the invalidation level, and decide whether the tape is accepting or rejecting the move.

References

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